

MARSHALL BRUNER

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EXPERIENCE

Ball Aerospace (DoD Clearance)

RF Engineer

Jun 2021 — Jul 2023

Broomfield, CO

- Designed and debugged multiple complex, mission-critical RF boards under strict time constraints
- Schematic capture and layout of RF T/R systems (RF, digital, power), antenna range test equipment, component testing coupons, etc.
- **Built data analysis and visualization tools in Python**
- Built an embedded webpage used to control/monitor phased array electronics
- **High sensitivity testing (noise, high power, etc.)** of electronics

Colorado State University

Graduate Research Assistant — Thesis focused on FMCW Radar

Jan 2021 — May 2022

Fort Collins, CO

- **Signal processing** / data visualization for Analog Device's **FMCW Phaser Radar**
- Design, schematic, and layout of a modular, X-band, FMCW drone radar
- Presented the work at the IEEE International Phased Array Symposium
- Phaser board work was presented by ADI at IMS 2022 and IEEE-APS 2022

CHILL Radar Lab

Lab Assistant

May 2020 — May 2021

Greeley, CO

- Built a server to host and display real-time and historical radar data using Post-GreSQL, Flask, Python
- Performed characterizations and measurements of radar equipment

EDUCATION

Colorado State University

M.S. Electrical Engineering — GPA: 3.9

Jan 2021 — May 2022

Fort Collins, CO

*Thesis - Design, Deployment, and Cost Considerations for DARMA; A Low-Cost and Lightweight **FMCW Radar***

Colorado State University

B.S. Electrical Engineering — GPA: 3.47

Aug 2017 — May 2021

Fort Collins, CO

PROJECTS

FMCW Radar Bootcamp

Mar 2023

- **Developed signal processing software and a presentation for an FMCW radar** to present at the 2022 AESS Radar Bootcamp

5G Electronics Senior Design Project

Jun 2020 — May 2021

- **Simulated, assembled, and measured a full 5G receiver system**
- Created teaching documentation for a 5G transmitter and receiver for FR1 and FR2 frequency bands
- Experience using FieldFox, EMPro, ADS, SystemVue

Fuzzy Logic Clutter Filter

Nov 2020 — Dec 2020

- **Designed a radar clutter filter using machine learning**

SKILLS & ABILITIES

Languages: C, Python, Rust, Bash, SQL, Matlab, NuShell, JavaScript

Design / Simulation Tools: AWR Microwave Office, HFSS, KiCad, AutoCAD, QucsStudio, LTSpice, Cadence Technologies / **Tools:** Benchtop measurement equipment, Numpy, Pandas, Scipy, Git, L^AT_EX, Linux

Communication: Worked with customers through pre- and post-sales phases, Excellent Technical Communication and Presentation Skills, Experience Presenting PDR/CDRs, Technical Presentation Formatting